~~Alternative~~ bridge concept: from LHC to ~~μCollider~~

Some tentative thoughts before discussions/toward White Paper

K. Piotrzkowski (AGH University, Kraków)

Preamble

- HL-LHC stops in 2041 (or later) and there is **no real need** to decide now very long-term future of CERN
- **In 10 years** (or more) one can make proper decision
- However, *Phase One* LHeC at 20 GeV should be decided **very soon**, and that is compatible with **both** "bridge scenarios" – towards FCC or μCollider
- LHμC as "demonstrator" for μCollider/**part of its infrastructure** (with LHeC)

Ultimate bridge concept: **from LHC to Next Collider**

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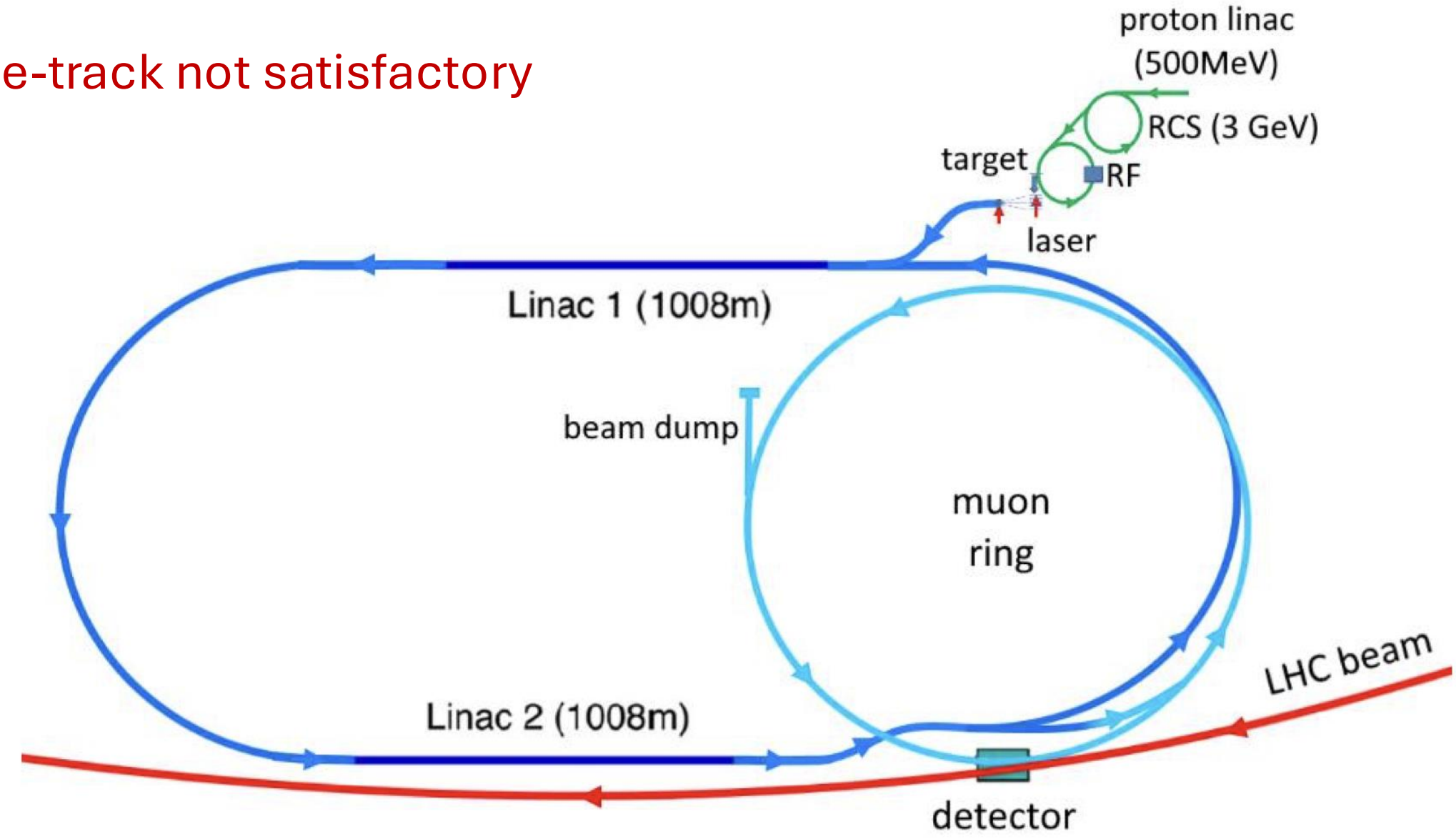
- LHμC as "demonstrator" for μCollider/part of its infrastructure (with LHeC); or, as "demonstrator" for **FCC-μh** at **10 TeV at least!** and ≈ symmetric!

LHμC: Alternative concept?

If race-track not satisfactory



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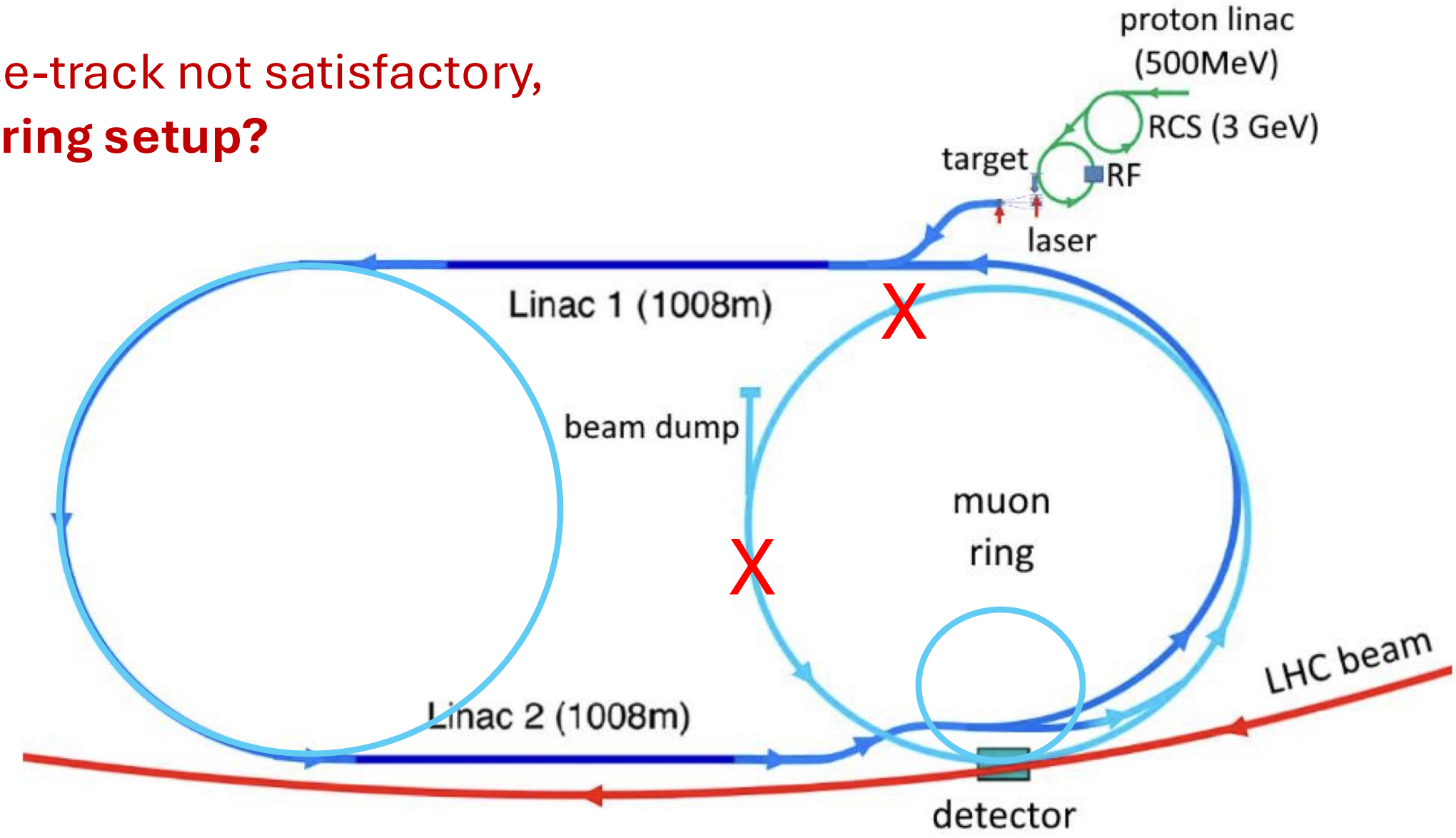

LHμC: Alternative concept?

If race-track not satisfactory, **ring-ring setup?**

- 430 GeV muons still provide μp at 3.5 TeV – as FCC-eh!
  and very symmetric!

- Small storage ring (1/4?) would improve luminosity (×4) , if bunches could be stacked